Worksheet A

Algebra and Functions

Edexcel International GCSE Mathematics Grades 6–8

Name: _____ Date: _____ 45 minutes 45 marks

Calculator allowed. Show all your working. Diagrams are not drawn accurately.

1 **[3]**

Solve $3x^2 - 7x - 6 = 0$

2 **[3]**

Simplify fully $\dfrac{x^2 - 9}{2x^2 + 5x - 3}$

3 **[5]**

Solve $\dfrac{2}{x+1} + \dfrac{3}{x-2} = 1$

Give your solutions in the form $a \pm \sqrt{b}$.

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4 **[7]**

$f(x) = \dfrac{x+4}{x-3} \quad x \neq 3 \qquad g(x) = 2x - 1$

(a) Find $f(5)$. [1]

(b) Find $f^{-1}(x)$. [3]

(c) Solve $fg(x) = 3$. [3]

5 **[5]**

Solve the simultaneous equations

$$y = 2x - 1$$
$$x^2 + y^2 = 10$$

6 **[3]**

$v^2 = u^2 + 2at$. Make t the subject of the formula.

7 **[3]**

Solve the inequality $x^2 - 5x - 14 > 0$

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8 **[3]**

Prove that $(2n+1)^2 - (2n-1)^2$ is a multiple of 8 for all integers n.

9 **[3]**

Here are the first four terms of a quadratic sequence: 5, 12, 23, 38, ...
Find an expression for the nth term.

10 **[4]**

y is inversely proportional to the square of x. When $x = 3$, $y = 4$.
(a) Find y when $x = 6$. [2]

(b) Find the positive value of x when $y = 9$. [2]

11 **[3]**

(a) Write $x^2 + 6x - 7$ in the form $(x + a)^2 + b$. [2]

(b) Write down the coordinates of the minimum point of $y = x^2 + 6x - 7$. [1]

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12 **[3]**

The graph of $y = f(x)$ has a turning point at $(2, -3)$. Write down the coordinates of the turning point of the graph of

(a) $y = f(x) + 4$ [1]

(b) $y = f(x - 1)$ [1]

(c) $y = -f(x)$ [1]

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